

Important Questions: BMC102 (Problem Solving Using C)

UNIT -I

1. Discuss the various approaches to problem solving.
2. Explain the various phases of SDLC(Software Development Life Cycle) or PDLC(Program Development Life Cycle)
3. What is an Algorithm? Write down the characteristics of a good algorithm.
4. What is a flowchart? Explain the various symbols used in the flow chart.
5. What are the three main principles of Structured Programming?
6. What are the advantages of modular programming?
7. Discuss the salient features of C language.
8. Explain the basic structure of a C Program with a suitable example.
9. Discuss the various stages involved in executing a C program.
10. What is a Token? Discuss the various types of tokens available in C language.
11. What are the rules of naming an Identifier in C?
12. Explain the working of various operators in C language using suitable examples with special emphasis on Logical, Bitwise and Increment/Decrement operators.
13. Discuss the various data types available in C language. Write a program to print the sizes of all the primitive data types in C.
14. Write down the precedence table of operators along with their associativity.
15. Explain the difference between Precedence and Associativity with suitable examples.
16. What is Type casting? Explain the difference between Implicit and Explicit typecasting with suitable example.
17. Explain the working of scanf and printf functions in C. Discuss the various format specifiers used with these functions.

UNIT -II

1. Explain the working of if-else if-else ladder using diagram and suitable example.
2. Write a program to print the grades of students in class using if-else if-else.
3. Write a program to print the electricity bill using if-else if-else.
4. Explain the working of switch-case control statement through an example. Also explain utility of break and default keywords in switch-case.
5. Write a program to check whether a given year is a leap year or not using if-else if-else and ternary operator through switch-case control statement.
6. Write a program to implement a basic arithmetic calculator using switch-case.
7. What are the different kinds of loops available in C language? Explain their working with suitable examples.
8. Explain the working of break and continue keywords in a nested-loop example.
9. Explain the working of goto keyword.
10. Write a program to print prime numbers in a given range.
11. Write a program to print perfect numbers in a given range.
12. Write a program to print Armstrong numbers in a given range.
13. Write a program to reverse a given number.
14. Write a program to print the sum of the digits of a given number.
15. What is a function? Discuss the various components of a function? Explain the following in context of functions: function declaration or function prototype, function definition and function calling.
16. Discuss the various categories of functions with suitable examples.
17. Illustrate the concept of recursion. Which data structure is used by recursion? What are the advantage and disadvantage of using recursion? Also discuss the types of recursion.
18. Write programs using Iteration and Recursion through switch-case for the following:
 - a. Factorial of given number

- b. GCD of two numbers
 - c. x raised to the power y (x^y)
 - d. Printing Fibonacci series
 - e. Sum of first n natural numbers
19. What is Call-by-value and Call-by-reference? Explain the difference through an example.
 20. Write a program to swap the values of two variables using call-by-value and call-by-reference.
 21. Can a function return more than one value at a time? Justify your answer with example.

UNIT -III

1. What is an array? Explain the importance of base address in an array with an example.
2. What is a pointer? Explain the difference between * and & operators in context of pointers.
3. What the different arithmetic operations possible with pointers? Write a program to implement it using an array through switch-case.
4. Is array a pointer in C? Justify your answer with an example.
5. Write a program to print the elements of an array using pointers.
6. How do we pass an array to a function? Explain with an example.
7. Write a program to implement Linear or Sequential search in C.
8. Write a program to copy the contents of an array to another.
9. Write a program to find the largest and the smallest element in an array.
10. Explain the meaning of a[i], &a[i], a+i, *(a+i), *a+i, i[a] in context of an array.
11. Explain the concept of Array of Pointers, Function Pointers and Pointer to Pointer with suitable example.
12. How a multidimensional array is declared, initialized and accessed.
13. Write a program to implement an array of variable size.
14. Write programs for the following:
 - a. Transpose of a Matrix
 - b. Addition and Subtraction of two matrices
 - c. Multiplication of two matrices of different orders
 - d. Sum of the elements on the Principal diagonal(Top Left to Right Bottom) of a Square matrix
15. What is a String in C? How do we declare, initialize and access elements of a String in C?
16. Explain the working of the following string manipulation functions in C: strlen(), strcpy(), strcat(), strcmp(), strrev().
17. Write a program to implement the working of the above string functions.
18. Write a program to reduce a multi-space string into a single-space string. Continuous multiple spaces must be reduced to a single space and if there is a single space between two words, it should remain as it is.
19. Illustrate the different ways to input a string in C language. Write a program to count the numbers of words and number of characters in an input string.
20. How a string is passed to a function. Explain with an example.
21. Write your own implementation of the following library functions : strlen(), strcpy(), strcat(), strcmp(), strrev()
22. Illustrate how array of Strings is represented in C.
23. Write a program to reduce a multi-space string into a single space string and then count the number of words in it.
24. Explain the concept of Array of String Pointers with suitable example. What is its advantage and what is its limitation?
25. Explain the concept of Pointer to a function with an example.
26. If given int y; int *ptr; Explain the meanings of the following expressions:
 - a. `y=*ptr++;`
 - b. `y=*++ptr;`
 - c. `y=++*ptr;`

d. `y=(*ptr)++;`

27. Explain the meanings of the following:

- a. `Int p[5];`
- b. `int *p[5];`
- c. `int (*p)[5];`
- d. `int *p();`
- e. `int (*p)();`

UNIT -IV

1. What is a structure in C? How is it different from an array? Write a program to declare, initialize and access a structure in C.
2. Explain array of structures with an example.
3. Explain with an example how structures can be nested
4. How can we pass a structure to a function? Give example.
5. What is a structure pointer? Explain the difference between `.`(dot) and `->`(arrow) operator with an example.
6. Explain the difference between Structure and Union.
7. Write a C program to store employee details such as Empid, Name, and Age for 50 employees and display the employee details who are getting salary more than 15000.
8. Write a C program to store 50 players' details such as Player Name, Country Name and Batting Average. Using the database display the details of the players who are playing for a given country.
9. Write a short note on enumerated data type in C.
10. What is the advantage of using Storage classes? Discuss the various storage classes in C with proper examples.

UNIT -V

1. What is Static and Dynamic memory allocation? What are the drawbacks of static memory allocation? Explain with the help of an example the working of various Dynamic memory allocation functions available in C.
2. What is a File? Explain the concept of streams in C.
3. Explain different categories of preprocessor directives used in C.
4. Write a program using nesting of macros to find the largest and smallest among three numbers.
5. How one can create our own header file and use it in our C program? Explain the difference between `#include<stdio.h>` and `#include "stdio.h"`. i.e. angle bracket approach and double quotes approach.
6. Discuss the various file opening modes in C with examples.
7. Explain the following file handling functions with suitable examples: `fopen()`, `fclose()`, `fgetc()`, `fputc()`, `fgets()`, `fputs()`, `feof()`, `fscanf()`, `fprintf()`, `fread()`, `fwrite()`.
8. What is Random access to a file? Discuss the random access file handling functions with examples: `fseek()`, `ftell()` and `rewind()`.
9. What are command line arguments? Explain with an example the process of running a program using command line arguments.
10. Write a program to find the sum of two numbers passed as command line arguments.(Use the function `atoi()` to convert string to int)
11. Write a program to display the contents of a file on the screen using command line arguments.
12. Write a file-copy program using command line arguments in textual and binary mode.
13. Explain the following graphics library functions for drawing and filling: `initgraph()`, `rectangle()`, `circle()`, `line()`, `ellipse()`, `fillellipse()`, `drawpoly()`, `fillpoly()`, `outtext()`, `outtextxy()`, `getmaxx()`, `getmaxy()`, `setfillstyle()`, `setcolor()`, `setbkcolor()`, `moveto()`.
14. Write a graphics program in C to display a red circle inside a blue square with all its sides being tangent to the circle.

15. Write a program to display 50 concentric circles of different colors in the centre of the screen .
16. How GUI interaction is done in a graphics program?